

MULAKKAL ROSEMOL SAJU

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Aspiring cybersecurity professional and undergraduate currently pursuing a BSc in Information and Communication Technology, complemented by a strong foundation in cybersecurity and digital forensics. Brings hands-on experience in incident response, vulnerability assessment, digital forensics, and software testing. Skilled in identifying security weaknesses, analysing system artefacts, and producing structured security reports using industry tools and frameworks. Highly motivated to contribute to national cybersecurity operations while continuously developing technical and analytical skills.

Education

- **Singapore University of Social Sciences** | 2025 – 2029
 - **BSc in Information and Communication Technology** (*In Progress*)
- **Temasek Polytechnic** | 2024 – 2026
 - **Diploma in Cybersecurity & Digital Forensics**
 - *Note: Transitioned to BSc-ICT following the completion of 3 semesters.*
- **Yishun Innova Junior College** | 2022 – 2023
 - **GCE A Levels (PCME)**

Experience

Trainer | Zenitant 2026 – *Present*

- Conducted and delivered technical workshops on computational thinking, micro:bit programming, and artificial intelligence fundamentals (Code for Fun / AI for Fun) to primary and secondary school students.
- Translated complex technical concepts into clear, accessible language while managing classroom engagement and instructional materials.
- Coordinated with program leads to ensure seamless execution of training modules and session delivery.

Software Tester (Freelance, Remote) — Testlio / OKX Sept 2025 -*Present*

- Conducted functional and regression testing for a large-scale fintech mobile application on Android and iOS.
- Verified application flows, usability, and localisation accuracy.
- Identified workflow and security-relevant issues, contributing to improved application reliability and risk reduction.
- Documented defects clearly for engineering teams and participated in regression validation.

Junior Software Tester — Guud Company *Dec 2021 – Jan 2022*

- Executed structured test cases to identify functional defects and edge-case behaviours.
- Documented bugs and collaborated with project stakeholders to improve software quality.
- Strengthened analytical skills in system behaviour, anomaly detection, and quality assurance.

Scholar — Girls Who Code *Apr 2024 – Present*

- Completed structured training in cybersecurity, data structures, machine learning, and web development.
- Applied technical concepts through hands-on projects simulating real-world challenges.

Skills & Abilities

- **Cybersecurity:** Incident Response, Threat Analysis, Vulnerability Assessment, Digital Forensics, ISO 27001
- **Tools:** Kali Linux, Burp Suite, SQLMap, Nmap, FTK Imager, Autopsy, Cisco Packet Tracer, PowerBI, KNIME
- **Programming & Scripting:** Python, Java, SQL, HTML, CSS
- **Systems & Networking:** Linux, VLANs, OSPF, NAT, SSH Hardening, Zero Trust Principles
- **Testing & QA:** Test Case Execution, Bug Reporting, Regression Testing
- **UX/UI:** Usability Testing, Interface Evaluation

Cybersecurity Projects

Incident Response & Ethical Hacking Case Study

- Performed penetration testing using Kali Linux, Burp Suite, SQLMap, and Nmap.
- Identified and exploited vulnerabilities including SQL injection, XSS, and privilege escalation.
- Produced a professional incident report with CVSS scoring, impact assessment, and mitigation strategies, simulating real-world response workflows.

Digital Forensics Investigation

- Conducted forensic acquisition and analysis using FTK Imager and Autopsy.
- Recovered deleted artefacts, analysed browser activity, and constructed forensic timelines.
- Applied evidence handling best practices and documented findings in a structured investigation report.

Enterprise Network Security Design

- Designed and secured a multi-VLAN enterprise network with DHCP, OSPF, NAT, and email services.
- Implemented SSH-only access, system hardening, and Zero Trust principles.
- Proposed improvements for resilience, monitoring, and failover.

E-Smart Vending Machine Web Application (Full Stack)

- Built a full-stack web application using Flask (Python), HTML/CSS/JavaScript, and MySQL.
- Implemented authentication, order handling, admin functions, and real-time updates.
- Focused on secure backend logic, database integration, and user-centric UI/UX.

Certifications & Activities

- AWS Certified Cloud Practitioner
- Google Cybersecurity Specialization
- CS50 Introduction to Cybersecurity (HarvardX)
- ISC2 Incident Response & Security Principles
- Cisco Networking Academy – Basic Networking
- L1: Basic Proficiency in KNIME Analytics Platform
- OWASP Hackathon Participant
- Girls in Tech x NCS Hackathon Participant

Volunteering & Leadership

- **Student Volunteer (Secretary)** — National Kidney Foundation
- **Teacher (Primary 5)** — Church of the Holy Trinity
- **Catechist** — Syro Malabar Church of Singapore
- **Research Volunteer** — Zooniverse (Environmental Data Projects)